A General Cooperative Optimization Driven High-Frequency Enhancement Framework for Multispectral Image Fusion

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***Abstract*— Pan-sharpening essentially to boost the spatial resolution of a multispectral (MS) image guided by its paired panchromatic (PAN) image. In other words, this process intricately integrates the high-frequency components extracted from texture-rich PAN images into the low-resolution (LR) MS images, resulting in texture-rich MS images. Though existing deep learning (DL)-based techniques have made impressive performance compared with traditional algorithms, they still face challenges in accurately restoring high-frequency details in MS images, thus limiting overall pan-sharpening performance. In addition, reference high-resolution (HR) MS images are often underutilized, typically serving only as training labels. In this work, we present a general high-frequency enhancement framework for pan-sharpening, which is implemented through a cooperative optimization strategy using mutual information (MI) maximization and contrastive learning. Specifically, our model comprises two fundamental modules: the high-frequency feature alignment (HFFA) module and the high-frequency detail calibration (HFDC) module. The first employs MI maximization to align the high-frequency semantic statistical distribution between PAN images and reference HRMS images. The latter is designed to calibrate the high-frequency components of MS modality under the guidance of the PAN counterparts through the contrastive learning constraint, thereby producing more accurate high-frequency information on MS modality. By integrating the calibrated high-frequency features of MS modality and those of PAN modality, we can obtain a more comprehensive and precise high-frequency feature representation of these two modalities, facilitating the reconstruction of LRMS images. Our model, incorporating the aforementioned key elements, significantly surpasses other state-of-the-art (SOTA) techniques across multiple satellite datasets in both quantitative and qualitative experiments. Moreover, the real-world full-resolution and cross-sensor assessments testify to its exceptional generalization capabilities. The code is available at https://github.com/Vcocoi/CONet.**

***Index Terms*— Contrastive learning, high-frequency enhancement, mutual information (MI) maximization, pan-sharpening.**

I. Introduction

PAN-SHARPENING is an image fusion technique to synthesize a desired high-resolution multispectral (HRMS) image from a low-resolution multispectral (LRMS) image and its paired high-resolution (HR) panchromatic (PAN) image. Due to hardware limitations on satellites, it is challenging to obtain remote sensing images that offer both HR and rich spectral information simultaneously. As a result, pan-sharpening has garnered significant interest from both the computer vision and remote sensing communities. Recently, the rapid development of deep learning (DL) technology has substantially advanced the field of pan-sharpening. PNN [1] is the first to introduce the convolutional neural network (CNN) to the field of pan-sharpening, yielding promising performance using a simple three-layer CNN architecture. Since then, numerous CNN-based pan-sharpening methods have emerged, such as PanNet [2], DiCNN [3], GPPNN [4], and FusionNet [5], among others, demonstrating the powerful learning capabilities of CNNs. In addition, SFIIN [6] is the pioneering attempt to address pan-sharpening in both frequency and spatial domains. MutNet [7] introduces the mutual information (MI) minimization to encourage the cross-modal complementary information learning. BiMPan [8] explores the local specificity of each spectral band in MS modality and consequently puts forward a bidomain modeling framework tailored for global and local learning.

Delving into the attributes of the pan-sharpening task, it is an image super-resolution process that injects PAN details into LRMS images. The key challenge in this process is to accurately reconstruct the high-frequency details in the MS modality. Very recently, several high-frequency enhancement techniques for pan-sharpening have been proposed to address this issue, demonstrating notable performance improvements. Zhou et al. [9] propose a framework by adaptively learning low-high frequency information integration in both frequency and spatial domains. Meng et al. [10] devise a coarse-to-fine detail restoration framework via band-limited multiscale learning. Wang et al. [11] apply an

Received 14 August 2024; revised 31 October 2024 and 26 December 2024; accepted 10 January 2025. Date of publication 20 February 2025; date of current version 24 March 2025. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 62301151 and in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2021A1515110949. *(Chentong Huang and Junming Hou contributed equally to this work.) (Corresponding authors: Junling Li; Man Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2025.3541561

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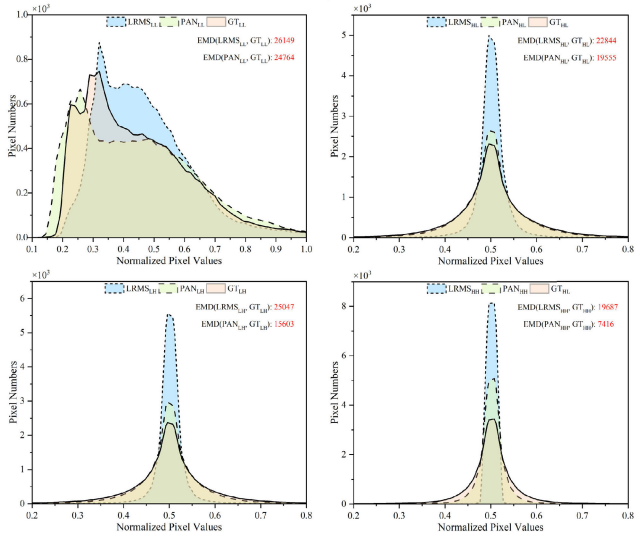


Fig. 1. Our motivation. The distribution statistics of different frequency components of LRMS images, PAN images, and GT images on a group of reduced resolution samples from the GaoFen-2 dataset. We also compute the EMD [15] between the distribution statistics of LRMS and PAN images relative to GT, offering a quantitative characterization of their discrepancies against GT, where a smaller EMD value suggests a higher similarity with GT. (Please see zoomed-in view for more details of each distribution pattern.)

implicit neural function to represent the high-frequency signals as functions of continuous coordinates to address misalignment issues. *However, these methods tend to roughly fuse high-frequency components of PAN and LRMS images, overlooking cross-modality mismatch. This intrinsic inconsistency between PAN and MS modalities may potentially introduce spatial and spectral distortions, degrading the overall pan-sharpening performance* [12], [13], [14]. To this end, we propose a general cooperative optimization-driven high-frequency enhancement framework to achieve precise detail reconstruction. To be specific, rather than merely using the reference HRMS images as training labels like most previous works, we leverage MI maximization between the reference HRMS images and PAN images in the high-frequency components to ensure cross-modal high-frequency consistency between MS and PAN modalities. In this manner, PAN images, as commonly used guidance for the super-resolution of LRMS images, provide more accurate HR details in assisting the detail preservation of LRMS images. Subsequently, we apply a contrastive learning constraint between PAN and MS modalities under the optimized PAN guidance to calibrate inaccurate cross-modal high-frequency information, with the aim of efficiently reconstructing authentic texture details.

A. Our Motivation

To explicitly shed light on the disparities in high-frequency components between the LRMS and PAN images compared with the reference HRMS images [used as ground truth (GT)], we take a group of source images from the GaoFen-2 dataset as an example and analyze the distributions of different frequency components decomposed by discrete wavelet transform (DWT). As depicted in Fig. 1, it is evident that the statistical distributions of the high-frequency components of LRMS images vary significantly from those of GT images, whereas those in PAN images closely match GT images. Furthermore, we compute the Wasserstein distance, also known as the Earth mover's distance (EMD), between the statistical distributions of LRMS and PAN images compared with GT images to quantitatively characterize their disparities in high-frequency components. Similarly, the PAN images demonstrate smaller EMD values than LRMS images, affirming a higher resemblance to GT images.

Based on the aforementioned observations, we devise two core building designs to obtain accurate high-frequency information. First, we devise a high-frequency feature alignment (HFFA) module implemented with MI maximization to better match the GT images in the high-frequency components. Instead of solely treating the GT images as training labels like the previous works, we leverage them as guidance for feature learning. Second, unlike previous approaches that roughly merge the high-frequency details of LRMS and PAN images without considering the inaccuracies inherent in the LRMS images, such as blurry texture edges, we propose a high-frequency detail calibration (HFDC) module. To be specific, the HFDC module is designed to calibrate the inaccurate high-frequency components of MS modality under the guidance of the PAN counterparts through the contrastive learning constraint. Through the dual-level guidance provided by GT images and PAN images, our cooperative optimization-driven framework refines cross-modal high-frequency information, thereby facilitating a more comprehensive and accurate reconstruction of the texture-rich HRMS images. We conduct extensive experiments over multiple satellite datasets to validate the efficacy of our model. Both qualitative and quantitative results demonstrate that our method significantly outperforms other cutting-edge techniques, particularly showcasing exceptional generalization capability.

Overall, our contributions can be summarized as follows.

1) Unlike previous works that only use GT images as training labels, we introduce an HFFA module implemented with MI maximization to constrain the high-frequency consistency between PAN images and GT images, since the former are commonly used as the HR guidance to spatially super-resolve the LRMS.
2) Considering the inaccurate high-frequency details of LRMS images, we devise an HFDC module through contrastive learning constraints, aiming to calibrate the high-frequency information of MS modality under PAN guidance refined by the HFFA module.
3) Extensive experiments on multiple remote sensing satellite datasets demonstrate the quantitative and qualitative superiority of our model against state-of-the-art (SOTA) methods.

The remainder of the article is organized as follows. In Section II, the related works are presented. In Section III, the proposed pan-sharpening framework will be introduced in detail. Experimental results and related discussions are presented exclusively in Section IV. Finally, the conclusion is drawn in Section V.

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II. Related Works

A. Traditional Pan-Sharpening Methods

Pan-sharpening techniques can be categorized into traditional methods and DL-based methods. Traditional methods are broadly classified into three types, component substitution (CS), multiresolution analysis (MRA), and variational optimization (VO). In CS methods [16], [17], [18], LRMS images are decomposed into spectral information and spatial textures. The spatial textures of the LRMS image are then substituted with corresponding details from the PAN image to generate the HRMS image. The most often used approaches in CS methods are intensity–hue–saturation (IHS) fusion [19], the principal component analysis (PCA) methods [20], Brovey transforms [21], and Gram–Schmidt (GS) orthogonalization methods [22]. These approaches accurately retain the spatial characteristics of PAN images, but they often neglect the spectral information of PAN images and the spectral misalignment between the two modal images, resulting in severe artifacts. Compared with CS methods, MRA-based approaches [23], [24], [25] contain less spectral distortion. MRA-based methods decompose the LRMS image over multiresolution to extract its spatial components, and then replace them with the high-frequency details from the PAN image. Representative MRA methods include the high-pass filter fusion (HPF) [25], generalized Laplacian pyramid (GLP) [26], and Atrous Wavelet Transform (ATWT) [24]. While these methods perform well in spectral information, multiresolution decomposition technology may introduce some spatial distortion. In contrast to the aforementioned two methods, VO methods [27], [28], [29] have emerged to alleviate spectral distortion and enhance spatial resolution. These VO methods aim to design an optimized function based on prior constraints and hypotheses. However, they may result in degradation due to inadequate characterization capabilities.

B. DL-Based Pan-Sharpening Methods

Over the past few years, the DL technology has greatly advanced various low-level vision tasks [30], [31], [32], [33], [34], [35], [36]. In the field of pan-sharpening, PNN [1] is the first effort that introduces a three-layer CNN, delivering competitive performance compared to traditional algorithms despite its simple architecture. Inspired by PNN, explosive CNN-based pan-sharpening techniques have been put forward to improve the learning capability of pan-sharpening [6], [7], [37], [38], [39]. PanNet [2] first introduces residual learning in pan-sharpening network design. SFIIN [6] is the pioneering work in pan-sharpening to exploit the frequency-domain information. MutNet [7] improves feature representation capability via MI minimization. Furthermore, GuidedNet [37] addresses pan-sharpening with fewer parameters and remarkable generalization capability. In addition to CNN-based methods, some Transformer-based networks [39], [40], [41] have also made significant advancements in the field of pan-sharpening. Very recently, with the success of diffusion models in various vision tasks, they have been introduced in pan-sharpening [38], [42], [43], demonstrating highly competitive performance, albeit with substantial computational costs. Despite significant advancements, high-frequency reconstruction remains a key challenge. On one hand, high-frequency texture details are often irreversibly lost during data simulation due to operations, such as blurring and downsampling [44], [45]. In addition, high-frequency details are often diverse and intricate, with their characteristics varying significantly across different scenes [8], [9]. This makes the accurate restoration of high-frequency components a critical factor in enhancing model generalization. Consequently, recent approaches propose diverse high-frequency enhancement strategies, demonstrating notable effectiveness [10], [11], [46]. However, these methods simply ignore the inaccurate detail information in LRMS modality, which significantly differs from the high-frequency components of GT counterparts, as shown in Fig. 1. To address this intrinsic limitation, we explore the potential of GT images to assist the high-frequency reconstruction of LRMS images, going beyond the conventional use as mere training labels in previous approaches.

C. Contrastive Learning

Recently, the contrastive learning has attracted comprehensive attention in computer vision for its powerful feature representation ability. Specifically, the contrastive learning constructs paired positive and negative samples and adjusts the distance between samples in the feature space, uncovering latent data correlation. Recent studies have applied contrastive losses in low-level vision tasks and achieved promising performance, e.g., infrared and visible image fusion [47], [48], image denoising [49], and image super-resolution [50]. However, there are still few works that apply contrastive learning in the field of pan-sharpening. Zhou et al. [51] introduce contrastive learning loss to effectively handle inconsistent information for feature distillation. Nevertheless, they lack the exploration of contrastive learning in handling multimodality disparities. As illustrated in Fig. 1, the high-frequency components of LRMS images exhibit significant discrepancies against GT images compared with those of PAN images. It may deteriorate the overall performance [12], [13]; yet, it has been rarely considered in previous works. To this end, we utilize contrastive constraints to calibrate inaccurate texture details in LRMS images, achieving effective multimodal detail restoration.

D. Mutual Information

InfoMax principle [52] improves the representation capability by maximizing the MI between the input and output. Since then, explosive works have utilized MI strategy for the effective representation in the computer vision community [7], [53], [54], [55], [56], [57]. For example, Zhang et al. [55] adopt MI minimization to explicitly encourage multimodality learning between RGB images and depth data. Sanghi [56] maximize the MI between 3-D objects and their geometrically transformed versions to enhance the representations. In the field of pan-sharpening, MutNet [7] reduces the cross-modality redundancy between PAN and LRMS by minimizing the MI, boosting the multimodality learning performance. Inspired by its success, we seek to leverage MI to alleviate the inherent

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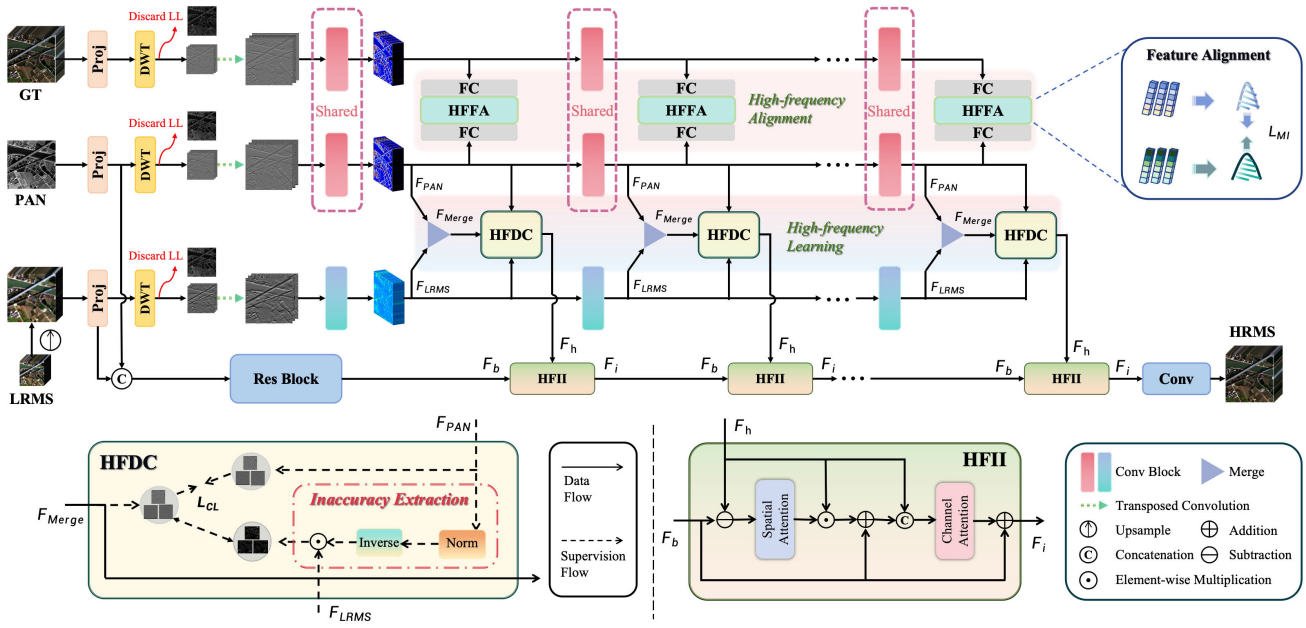


Fig. 2. Our overall architecture. The high-frequency components of the corresponding GT image, the PAN image, and the LRMS image are obtained via Haar Wavelet after being projected into the feature dimension. Then, high-frequency components of different modalities are, respectively, fed into cascaded convolutional layers to enhance feature representation from shallow to deep levels. One of the two core buildings HFFA is implemented between GT features and PAN features to achieve feature-level alignment under the supervision of GT. The other core design HFDC is employed to calibrate cross-modal high-frequency details via the guidance of PAN. Finally, the cooperatively optimized details are injected into base features in the customized HFII module.

detail discrepancies between PAN and GT images, as depicted in Fig. 1, a critical aspect frequently overlooked by previous methods. Whereas pan-sharpening is fundamentally integrates the high-frequency details of PAN images into LRMS images, thus accurately extracting high-frequency PAN information is crucial for generating ideal MS images. In this work, we propose aligning the high-frequency information between PAN images and reference GT images through MI maximization, thereby obtaining more precise high-frequency guidance to boost the reconstruction of LRMS images.

III. Proposed Method

In this section, we first introduce the preliminary of wavelet transform. Then, we present the overall framework, followed by a detailed explanation of the three core designs, including the HFFA module, HFDC module, and high-frequency information injection (HFII) module. Finally, the network optimization is introduced.

A. Preliminary of Wavelet Transform

Wavelet Transform [58], an efficient and simple analytical tool commonly used in the computer vision field, is used to decompose images into one low-frequency subband and three high-frequency subbands. Specifically, the transformation comprises four sets of filters, $\mathbf{LL}^{\mathbf{T}}$, $\mathbf{HL}^{\mathbf{T}}$, $\mathbf{LH}^{\mathbf{T}}$, and $\mathbf{HH}^{\mathbf{T}}$, where low-pass filter $\mathbf{L}$ and high-pass filter $\mathbf{H}$ are defined as follows:

$$\mathbf{L} = \frac{1}{\sqrt{2}}[11], \quad \mathbf{H} = \frac{1}{\sqrt{2}}[-11]. \tag{1}$$

The localized nature of wavelet transform allows for precise localization of edge features in both the spatial and frequency domains. As shown in (2), we apply level-1 Haar Wavelet Transform to the input images, obtaining three high-frequency components, $\mathbf{X}^{\mathbf{HL}}$, $\mathbf{X}^{\mathbf{LH}}$, and $\mathbf{X}^{\mathbf{HH}}$, which represent the horizontal, vertical, and diagonal components, respectively, along with a low-frequency components, $\mathbf{X}^{\mathbf{LL}}$. According to (3), we concatenate the three high-frequency components along the channel dimension as the high-frequency inputs, while $\mathbf{X}^{\mathbf{LL}}$ is omitted from further network processing. This process can be expressed as follows:

$$\mathrm{DWT}(\mathbf{X}) = \mathbf{X}^{\mathbf{LL}}, \mathbf{X}^{\mathbf{HL}}, \mathbf{X}^{\mathbf{LH}}, \mathbf{X}^{\mathbf{HH}} \tag{2}$$

$$\mathbf{X}^{\mathbf{H}} = \mathrm{Cat}\left(\mathbf{X}^{\mathbf{HL}}, \mathbf{X}^{\mathbf{LH}}, \mathbf{X}^{\mathbf{HH}}\right) \tag{3}$$

where $\mathrm{DWT}(\cdot)$ denotes the DWT operation and $\mathrm{Cat}(\cdot)$ represents the concatenation operation along the channel dimension.

B. Overall Framework

As depicted in Fig. 2, we start by upsampling the LRMS image to match the size of the PAN image. Subsequently, we project the GT, PAN, and the upsampled LRMS images into the feature space using convolutional layers coupled with the nonlinear activation. After that, we apply the Haar wavelet transform to extract their high-frequency components, followed by transposed convolutions to restore the size to match the original dimensions prior to the Haar wavelet transform. Following this, we employ two key procedures based on our observations in Fig. 1, i.e., the high-frequency alignment and the high-frequency learning, to obtain more informative and accurate high-frequency details. Finally, by integrating the gradually refined high-frequency information into the base cross-modal features extracted by residual blocks (which can be substituted with any end-to-end pan-sharpening network), our method can effectively generate the desired HRMS images.

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C. Core Building Designs

1) High-Frequency Feature Alignment: Pan-sharpening fundamentally aims to effectively and precisely inject high-frequency information from PAN images into LRMS images [11], [46], [59]. Previous high-frequency enhancement methods directly integrate the high-frequency information from PAN images with LRMS images [10], [11], [46]. On the one hand, they simply ignore the inherent discrepancies between PAN and GT images in high-frequency components despite being shown in Fig. 1. On the other hand, there is limited exploration of GT images, which are merely used as labels. Consequently, we propose aligning the high-frequency information of source PAN images under the guidance of GT images, acquiring more accurate high-frequency details to assist the reconstruction of LRMS images. Technically, we maximize MI between the high-frequency features of PAN and GT images to achieve this goal. MI is a fundamental concept in information theory that quantifies the information shared between two random distributions. According to the information theory, the MI between random distributions $\mathbf{M}$ and $\mathbf{N}$ can be expressed as follows:

$$\mathrm{MI}(\mathbf{M},\mathbf{N}) = H(\mathbf{M}) + H(\mathbf{N}) - H(\mathbf{M},\mathbf{N}) \tag{4}$$

where $H(\cdot)$ denotes the function of marginal entropy and $H(\cdot,\cdot)$ is the function of joint entropy. Furthermore, we can formulate the MI using Kullback–Leibler (KL) divergence as follows:

$$\mathrm{MI}(\mathbf{M},\mathbf{N}) = D_{\mathrm{KL}}(P(\mathbf{M},\mathbf{N})||P(\mathbf{M})P(\mathbf{N})) \tag{5}$$

where $D_{\mathrm{KL}}(\cdot)$ denotes the KL divergence. $P(\cdot)$ and $P(\cdot,\cdot)$ represent the marginal probability distribution and joint probability distribution, respectively.

While MI stands as a theoretically concise and elegant measure, its computational poses significant challenges. This is primarily due to the intricate task of accurately calculating both the marginal and joint distributions simultaneously. Thus, the majority of research endeavors opt to estimate this quantity instead. Previous works [60], [61] have proved that it's feasible to substitute KL divergence with Jensen–Shannon (JS) divergence, effectively addressing the asymmetry of KL divergence

$$D_{\mathrm{JS}}(\mathbf{X}||\mathbf{Y}) = \frac{1}{2}D_{\mathrm{KL}}\left(\mathbf{X}||\frac{\mathbf{X}+\mathbf{Y}}{2}\right) + \frac{1}{2}D_{\mathrm{KL}}\left(\mathbf{Y}||\frac{\mathbf{X}+\mathbf{Y}}{2}\right) \tag{6}$$

where $\mathbf{X}$ and $\mathbf{Y}$ represent the two random probability distributions. $D_{\mathrm{JS}}(\cdot)$ denotes the function of JS divergence. Experimentally, replacing KL divergence with JS divergence proves to be stable in the training stage. Hence, we introduce JS divergence to estimate MI for maximizing the MI with no need for precise values of MI. As clearly illustrated in Fig. 2, the GT image and the PAN image are first projected into the feature space through shallow convolutional layers, followed by the wavelet transform for high-frequency component extraction. After being restored to their original sizes through transposed convolution operations, the high-frequency features of GT and PAN are then progressively processed through shared convolutional layers for deeper feature representations. As depicted in the high-frequency alignment branch, these high-frequency features of GT and PAN are further concurrently converted into vectors via fully connected layers at each stage to estimate the MI between them. Mathematically, we can write this process as follows:

$$\begin{aligned}
\mathbf{F}_{\mathbf{g}}^{\mathbf{H}} &= \phi^{\star}(T\mathrm{Conv}(\mathrm{DWT}(\mathrm{Proj}(\mathbf{G})))) \\
\mathbf{F}_{\mathbf{p}}^{\mathbf{H}} &= \phi^{\star}(T\mathrm{Conv}(\mathrm{DWT}(\mathrm{Proj}(\mathbf{P})))) \\
\mathbf{V}_{\mathbf{g}} &= \mathrm{FC}\left(\mathbf{F}_{\mathbf{g}}^{\mathbf{H}}\right), \quad \mathbf{V}_{\mathbf{p}} = \mathrm{FC}\left(\mathbf{F}_{\mathbf{p}}^{\mathbf{H}}\right) \\
\mathrm{MI}' &= D_{\mathrm{JS}}(P(\mathbf{V}_{\mathbf{g}},\mathbf{V}_{\mathbf{p}})||P(\mathbf{V}_{\mathbf{g}})P(\mathbf{V}_{\mathbf{p}}))
\end{aligned} \tag{7}$$

where $\mathbf{G}$ and $\mathbf{P}$ denote the input GT and PAN images, respectively. $\mathrm{Proj}(\cdot)$ indicates the process of projecting input images into the feature space, while $\phi^{\star}(\cdot)$ denotes the mapping of convolutional layers shared by PAN and GT features. $\mathrm{TConv}(\cdot)$ refers to the transposed convolutional operation, which restores the features to their original size prior to applying the DWT. $\mathbf{F}_{\mathbf{g}}^{\mathbf{H}}$ and $\mathbf{F}_{\mathbf{p}}^{\mathbf{H}}$ are the extracted high-frequency features of GT and PAN images, respectively. $\mathrm{FC}(\cdot)$ denotes the fully connected layer. $\mathbf{V}_{\mathbf{g}}$ and $\mathbf{V}_{\mathbf{p}}$ are the encoded feature vectors for MI estimation. MI' denotes the use of JS divergence as a substitute for traditional MI calculation between GT and PAN features, providing a basis for further optimization.

Afterward, we employ the Deep InfoMax [60] estimator to conduct the MI maximization. The JS divergence in (7) can be estimated as follows [60], [61]:

$$\begin{aligned}
\dot{D}_{\mathrm{JS}}^{\theta}(P(\mathbf{M})||P(\mathbf{N})) = {} & E_{p(\mathbf{m},\mathbf{n})}\left[-\log(1+e^{-T_\theta(\mathbf{m},\mathbf{n})})\right] \\
& - E_{p(\mathbf{m})p(\mathbf{n})}\left[\log\left(1+e^{T_\theta(\mathbf{m},\mathbf{n})}\right)\right]
\end{aligned} \tag{8}$$

where $\dot{D}_{\mathrm{JS}}^{\theta}(\cdot)$ denotes the estimated JS divergence and θ represents the optimizable parameters in the statistics network T_θ, $\mathbf{m} \in \mathbf{M}$ and $\mathbf{n} \in \mathbf{N}$. $E_{p(\cdot)}$ means the expectation value. By applying the D_{JS} divergence estimator, we can optimize the MI constraint between $\mathbf{V}_{\mathbf{g}}$ and $\mathbf{V}_{\mathbf{p}}$ as follows:

$$L_{\mathrm{MI}} = -\sum_{i=1}^{N} \alpha_i \dot{D}_{\mathrm{JS}}^{\theta}\left(P\left(\mathbf{V}_{\mathbf{g}}^{\mathbf{i}},\mathbf{V}_{\mathbf{p}}^{\mathbf{i}}\right)\middle|\middle|P\left(\mathbf{V}_{\mathbf{g}}^{\mathbf{i}}\right)P\left(\mathbf{V}_{\mathbf{p}}^{\mathbf{i}}\right)\right) \tag{9}$$

where i denotes the stage number. N is the number of the cooperative optimization stages, which is defaulted to 5 in this work. α_i is the weight of the ith stage MI loss. With the supervision of MI maximization between high-frequency features of GT and PAN images, we can capture desired PAN detail information to effectively guide the reconstruction of LRMS images. It is important to note that the high-frequency alignment procedure, implemented through the key HFFA module, is applied only during the training stage.

2) High-Frequency Detail Calibration: As depicted in Fig. 1, the high-frequency components of LRMS images exhibit considerable disparities against GT images compared to those of PAN images. To alleviate the inherent cross-modal inconsistency in the high-frequency distributions that may cause adverse effects during the fusion process [12], [13], we seek to calibrate the detail information of LRMS images under the guidance of high-frequency PAN features optimized by the HFFA module, thereby acquiring more consistent cross-modal detail features. To be specific, the upsampled

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LRMS images are first projected into the feature space with shallow convolutional layers. Then, DWT and additional convolutional operations, including transposed convolutions, are applied to obtain high-frequency LRMS features, $\mathbf{F}_{\mathbf{m}}^{\mathbf{H}}$. Subsequently, the optimized high-frequency PAN features and the high-frequency LRMS features are used to construct the positive samples $\mathbf{F}^{+}$ and negative samples $\mathbf{F}^{-}$, the process can be written as follows:

$$\begin{aligned} \mathbf{F}_{\mathbf{m}}^{\mathbf{H}} &= \phi(\mathrm{TConv}(\mathrm{DWT}(\mathrm{Proj}(\mathbf{M})))) \\ \mathbf{F}^{+} = \mathbf{F}_{\mathbf{p}}^{\mathbf{H}}, \quad \mathbf{F}^{-} &= \left(1 - \mathrm{Norm}\left(\mathbf{F}_{\mathbf{p}}^{\mathbf{H}}\right)\right) \cdot \mathbf{F}_{\mathbf{m}}^{\mathbf{H}} \end{aligned} \tag{10}$$

where M denotes the upsampled LRMS image. $\phi(\cdot)$ and $\mathrm{Norm}(\cdot)$ denote the mapping of convolutional layers and the function of normalization, respectively. In (10), normalized high-frequency features of PAN images are inversed as the kernels for filtering out the inconsistent high-frequency features in LMRS images. Furthermore, we merge the cross-modal high-frequency information of PAN and LRMS modalities to construct the target samples. Specifically, we first concatenate the extracted high-frequency PAN and LRMS features along the channel dimension, followed by shallow convolutional layers coupled with the nonlinear activation function. By minimizing the distance between the target and positive samples $\mathbf{F}^{+}$, while pushing the target away from negative samples $\mathbf{F}^{-}$, we can obtain consistent high-frequency cross-modality information. The merging process and mathematical optimization objective can be formulated as follows:

$$\mathbf{F}_{\mathrm{merge}} = \mathrm{Conv}\left(\mathrm{Cat}\left(\mathbf{F}_{\mathbf{p}}^{\mathbf{H}}, \mathbf{F}_{\mathbf{m}}^{\mathbf{H}}\right)\right) \tag{11}$$

$$L_{\mathrm{CL}} = \sum_{i=1}^{N} \beta_i \frac{\left|\left|\mathbf{F}_{\mathbf{merge}} - \mathbf{F}_{\mathbf{i}}^{+}\right|\right|_1}{\left|\left|\mathbf{F}_{\mathbf{merge}} - \mathbf{F}_{\mathbf{i}}^{-}\right|\right|_1} \tag{12}$$

where $\mathbf{F}_{\mathbf{merge}}$ denotes the merged cross-modal high-frequency features. β_i is the weight of the ith stage of contrastive loss and $||\cdot||_1$ represents L_1 loss.

3) High-Frequency Information Injection: After refining the cross-modal high-frequency features through the core HFFA and HFDC modules, we integrate them with base features to acquire more informative feature representations. As illustrated in Fig. 2, we first employ a vanilla residual block to extract the basic cross-modal features ($\mathbf{F_b}$) as follows:

$$\mathbf{F_b} = \mathcal{N}(\mathrm{Cat}(\mathrm{Proj}(\mathbf{P}), \mathrm{Proj}(\mathbf{M}))) \tag{13}$$

where $\mathcal{N}(\cdot)$ means the mapping of the residual block.

Next, we inject the optimized cross-modal detail information into the base features ($\mathbf{F_b}$) using an effective HFII module. As displayed in Fig. 2, the spatial attention [SA; with the mapping of $\mathrm{SA}(\cdot)$] is deployed after calculating the absolute difference between base features ($\mathbf{F_b}$) and refined high-frequency features ($\mathbf{F_h}$) for exploring interspatial dependencies. As detailed in Fig. 3, the input features (i.e., the residues between $\mathbf{F_b}$ and $\mathbf{F_h}$) first pass through a convolutional block. The obtained features are then processed through the global average pooling and global max pooling operations to aggregate spatial information. Subsequently, the outcomes from these two operators are concatenated along the channel dimension, followed by a convolutional block coupled with Sigmoid activation to generate the SA map which is applied to weight the original input features, producing the corresponding output. Finally, the output features of SA are further multiplied with $\mathbf{F_h}$ to enable texture compensations and then combined with the coarse-grained $\mathbf{F_b}$ via a residual connection. Formally, the above process can be formulated as follows:

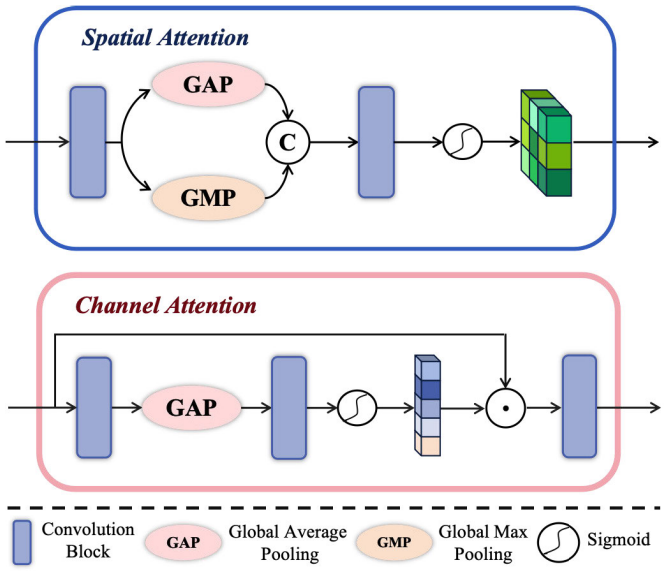


Fig. 3. Detailed structure of SA and CA within the HFII module.

$$\hat{\mathbf{F}}_{\mathbf{i}} = \mathbf{F_b} + \mathrm{SA}(\mathbf{F_b} - \mathbf{F_h}) \cdot \mathbf{F_h}. \tag{14}$$

To further enrich the feature representations, we introduce channel attention (CA), [with the mapping of $\mathrm{CA}(\cdot)$]. As depicted in Fig. 2, the output features $\hat{\mathbf{F}}_{\mathbf{i}}$ of SA are first concatenated with the extracted detail features $\mathbf{F_h}$, followed by the application of CA. The detailed structure of CA is shown in Fig. 3, where the input features are sequentially processed through a convolutional block, global average pooling, and another convolutional block, followed by sigmoid activation to generate the CA map. The input features are then weighted by the computed CA map, resulting in more informative feature representations, which are further refined through a convolutional block. Ultimately, the outcomes of CA is added with the base features $\mathbf{F_b}$ to produce the integrated features $\mathbf{F_i}$. This process can be mathematically represented as follows:

$$\mathbf{F_i} = \mathbf{F_b} + \mathrm{CA}\left(\mathrm{Cat}\left(\hat{\mathbf{F}}_{\mathbf{i}}, \mathbf{F_h}\right)\right). \tag{15}$$

D. Loss Function

The overall loss function consists of four parts

$$L = L_1 + \alpha L_{\mathrm{SSIM}} + \beta L_{\mathrm{MI}} + \gamma L_{\mathrm{CL}} \tag{16}$$

where α, β, and γ are the tradeoff parameters set to 0.1, 1, and 0.1, respectively, in the experiments. We adopt the $L1$ loss for implementing image reconstruction loss L_{rec}

$$L_{\mathrm{rec}} = ||\mathbf{H} - \widehat{\mathbf{H}}||_1 \tag{17}$$

where $\widehat{\mathbf{H}}$ and $\mathbf{H}$ denote the predicted MS images and GT images. In addition, the structure similarity loss L_{SSIM} is

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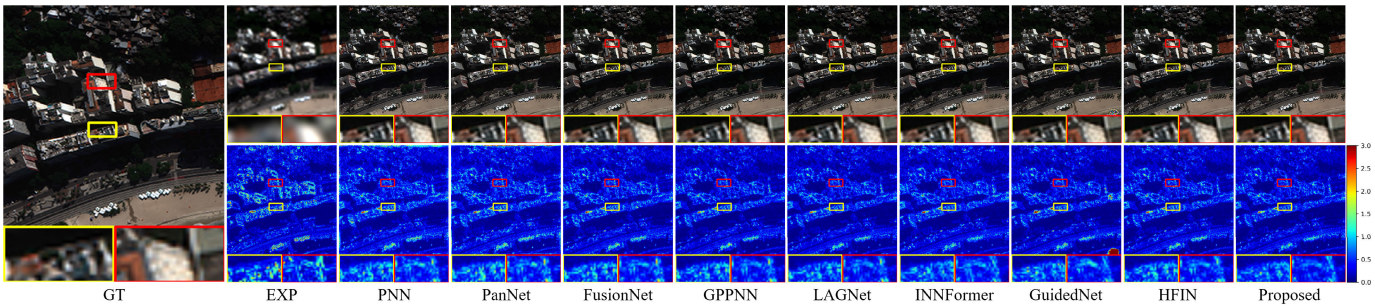


Fig. 4. Qualitative comparison on a reduced-resolution sample from WV3 dataset. The first row shows RGB visualizations, while the second row gives the corresponding absolute error maps between the pan-sharpened results and the GT image.

defined as follows:

$$L_{\mathrm{SSIM}} = 1 - S\left(\mathbf{H}, \widehat{\mathbf{H}}\right) \tag{18}$$

where $S(\cdot)$ is the mapping of SSIM function. The loss terms L_{MI} and L_{CL} are shown in (9) and (12).

IV. Experimental Details

In this section, we provide extensive experimental results conducted across three datasets. Our approach achieves SOTA performance, as confirmed by thorough qualitative and quantitative evaluations. In addition, ablation experiments validate the efficacy of each design within our proposed method.

A. Datasets and Benchmarks

1) Dataset Simulation: We experiment over multiple datasets including WorldView-3 (WV3), GaoFen-2 (GF2), and WorldView-2 (WV2), to validate the effectiveness of our method. Due to the absence of GT images, we leverage Wald's protocol [62] to simulate the training data, whose details can be referred to [44]. Each training set contains thousands of PAN/LRMS/GT image pairs with the sizes of $64 \times 64 \times 1$, $16 \times 16 \times s$, and $64 \times 64 \times s$, respectively. For the testing, there are 20 PAN/LRMS/GT image pairs with the sizes of $256 \times 256 \times 1$, $64 \times 64 \times s$, and $256 \times 256 \times s$ on the reduced-resolution evaluation, and 20 PAN/LRMS image pairs with the sizes of $512 \times 512 \times 1$ and $128 \times 128 \times s$ without GT images on the full-resolution assessment. s denotes the spectral band number which is 8 for WV3 dataset and 4 for GF2 dataset. In addition, the WV2 dataset sharing the same size as the WV3 dataset is used for cross-sensor generalization assessment. All training and testing data are available at.[1]

2) Quantitative Metrics: According to the previous research, five widely used image quality assessment (IQA) metrics, including PSNR, SAM [63], SCC [64], ERGAS [65], and Q-index [66] (Q4 for four-band datasets and Q8 for eight-band datasets), are adopted to perform the reduced-resolution assessment. In terms of the real-world full-resolution scenes, we use three no-inference metrics D_λ and D_s [14] and HQNR [67] as the IQA indexes for the absence of GT images.

[1]https://github.com/liangjiandeng/PanCollection

3) Baselines: To evaluate the effectiveness of our method, we select several SOTA pan-sharpening methods for comprehensive qualitative and quantitative assessments. Traditional algorithms: BT-H [21], BDSD-PC [68], MTF-GLP-HPM-R [69], and LRTCFPan [70]. DL-based methods: PNN [1], PanNet [2], FusionNet [5], GPPNN [4], LAGNet [71], INN-Former [41], GuidedNet [37], and HFIN [72].

4) Implementation Details: Our proposed model is implemented in the Pytorch framework with a batch size of 32 on a Linux operating system on a single NVIDIA RTX3090 GPU. To optimize the training process, we adopt Adam optimizer over 700 epochs with a dynamic learning rate to train the network, starting with an initial learning rate of 0.001, which gradually halves every 50 epochs.

B. Comparison With SOTA Methods

We conduct comprehensive experiments, including reduced- and full-resolution assessments as well as generalization investigation, on multiple datasets to illustrate the superiority over other SOTA methods.

1) Evaluation on Reduced-Resolution Scenes: We first implement the reduced-resolution evaluation to gauge the disparities between the predicted HRMS images and the reference GT images. Quantitative comparisons between our method and SOTA approaches on 20 testing samples, respectively, from WV3 and GF2 datasets are summarized in Table I. It is clear that our model outperforms other promising techniques across all reduced-resolution metrics, showcasing desired fusion performance. Notably, our method attains significantly higher PSNR scores, exceeding the second-best results by 0.4849 dB on the WV3 dataset and 0.7135 dB on the GF2 dataset. Similar positive trends are observed across other indicators, further validating the effectiveness of our proposed method.

We further qualitatively compare all DL-based approaches to further affirm the efficacy of our method. Figs. 4 and 5 offer the corresponding RGB visualizations over a representative sample from WV3 and GF2 datasets. It is evident that the products generated by our method preserve more clear detailed information as well as minimal spectral aberration. The last row of Figs. 4 and 5 further exhibits the residual maps between fused images and the corresponding reference GT images. In comparison with other leading pan-sharpening techniques, the residual map produced by our method exhibits fewer prominent bright spots, suggesting greater alignment with the

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TABLE I

QUANTITATIVE COMPARISON ON WV3 AND GF2 DATASETS. ↑ INDICATES THAT THE LARGER THE VALUE, THE BETTER THE PERFORMANCE AND ↓ INDICATES THAT THE SMALLER THE VALUE, THE BETTER THE PERFORMANCE. (RED: BEST AND BLUE: SECOND BEST)

Method	WorldView-3								Gaofen-2							
	Reduced					Full			Reduced					Full		
	PSNR↑	SAM↓	ERGAS↓	Q8↑	SCC↑	D_λ ↓	D_s ↓	HQNR↑	PSNR↑	SAM↓	ERGAS↓	Q4↑	SCC↑	D_λ ↓	D_s ↓	HQNR↑
EXP	29.1187	5.8001	7.1555	0.6270	0.7440	0.0232	0.0813	0.8975	32.1659	1.8200	2.3662	0.8123	0.8705	0.0209	0.0957	0.8881
BT-H	33.0796	4.9198	4.5789	0.8324	0.9252	0.0574	0.0810	0.8670	36.0541	1.6488	1.5280	0.9177	0.9570	0.0602	0.1313	0.8165
BDSD-PC	32.9690	5.4293	4.6976	0.8294	0.9081	0.0625	0.0730	0.8698	35.1800	1.6813	1.6667	0.8922	0.9521	0.0759	0.1548	0.7812
MTF-GLP-HPM-R	32.9808	5.3383	5.2301	0.8346	0.8892	0.0206	0.0630	0.9180	35.5576	1.6502	1.5875	0.8996	0.9480	0.0336	0.1404	0.8309
LRTCFPan	33.6134	4.7367	4.3153	0.8463	0.9273	0.0176	0.0528	0.9307	37.6072	1.2977	1.2723	0.9350	0.9637	0.0325	0.0896	0.8806
PNN	37.3093	3.6798	2.6819	0.8929	0.9761	0.0213	0.0428	0.9369	39.0712	1.0477	1.0572	0.9604	0.9772	0.0317	0.0943	0.8771
PanNet	37.3414	3.6156	2.6660	0.8906	0.9850	0.0165	0.0470	0.9374	40.2427	0.8749	0.7854	0.9538	0.9828	0.0179	0.0799	0.9036
FusionNet	38.0424	3.3252	2.4666	0.9044	0.9807	0.0239	0.0364	0.9406	39.6386	0.9735	0.9878	0.9641	0.9806	0.0350	0.1013	0.8673
GPPNN	38.5709	3.0554	2.3056	0.9143	0.9837	0.0281	0.0378	0.9352	42.4459	0.7972	0.7107	0.9791	0.9897	0.0229	0.0665	0.9122
LAGNet	38.5681	3.1042	2.2999	0.9098	0.9838	0.0368	0.0418	0.9230	42.7348	0.7859	0.6869	0.9804	0.9906	0.0284	0.0792	0.8947
INNFormer	38.3129	3.2174	2.3604	0.9117	0.9831	0.0550	0.0679	0.8815	42.8695	0.7875	0.6619	0.9801	0.9919	0.0609	0.1096	0.8360
GuidedNet	38.3129	2.9838	2.2711	0.9177	0.9844	0.0291	0.0410	0.9312	42.9552	0.8268	0.7191	0.9798	0.9901	0.0209	0.0573	0.9229
HFIN	38.5342	3.0876	2.3062	0.9116	0.9835	0.0249	0.0425	0.9338	42.1889	0.8427	0.7347	0.9774	0.9895	0.0272	0.0620	0.9124
Proposed	39.0558	2.9579	2.1706	0.9183	0.9858	0.0175	0.0344	0.9487	43.6687	0.6919	0.6196	0.9840	0.9923	0.0202	0.0472	0.9335

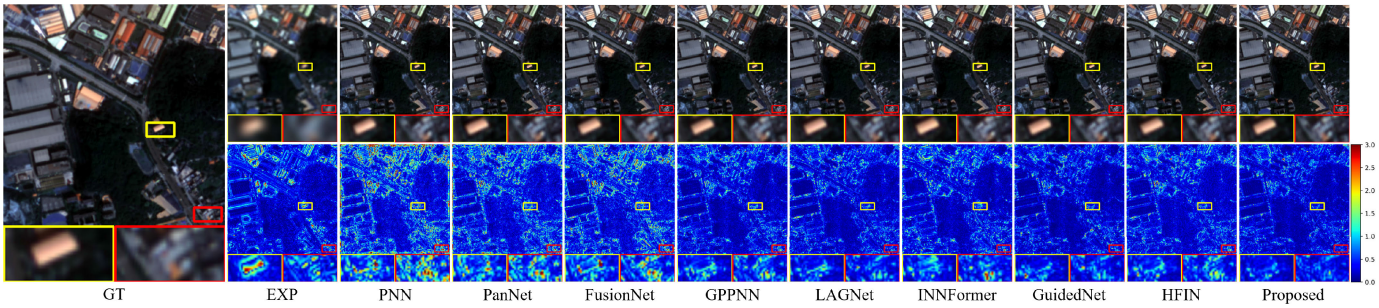


Fig. 5. Qualitative comparison on a reduced-resolution sample from GF2 dataset. The first row shows RGB visualizations, while the second row gives the corresponding absolute error maps between the pan-sharpened results and the GT image.

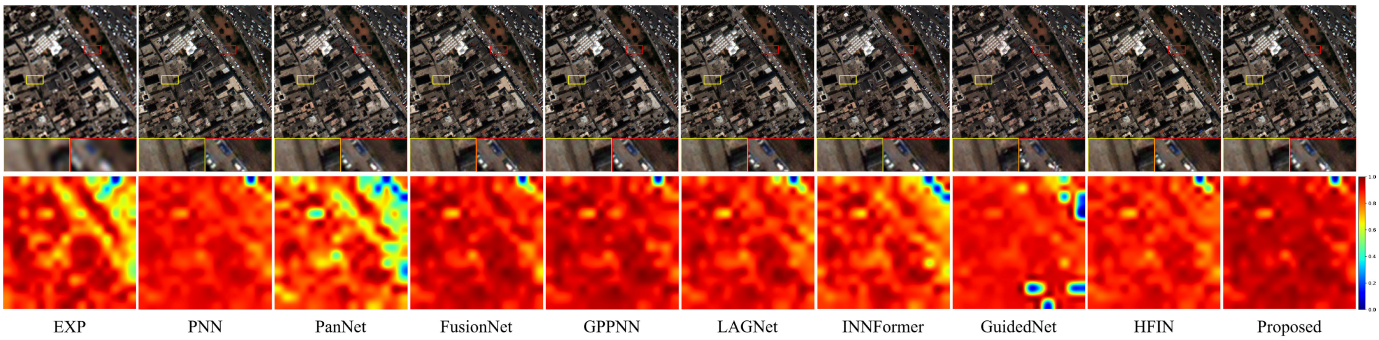


Fig. 6. Qualitative comparison on a full-resolution sample from WV3 dataset. The first row displays RGB visualizations, while the second row presents the corresponding HQNR maps. Darker colors indicate better pan-sharpening performance.

reference GT image. The consistent positive trends observed in both qualitative and quantitative findings strongly emphasize the superiority of our approach.

2) Evaluation on Full-Resolution Scenes: Full-resolution evaluation corroborates the generalization performance in real-world applications with no reference data. Table I displays the full-resolution quantitative comparisons between our method and other SOTA approaches on WV3 and GF2 datasets. It can be seen that our method surpasses other SOTA methods on the majority of all full-resolution indexes, demonstrating its favorable generalization performance. In particular, our model shows a significant improvement compared to the second-best method in the HQNR metric, indicating a notable advancement in overall pan-sharpening performance.

Regarding the qualitative results on the full-resolution, the first row of Figs. 6 and 7 presents the corresponding RGB visualizations of a typical sample from WV3 and GF2 datasets, respectively. Clearly, our model produces images with

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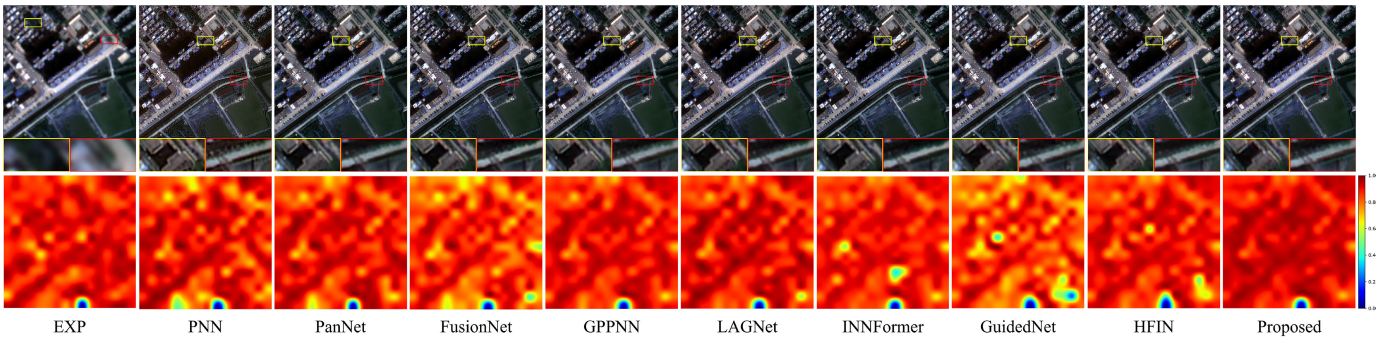


Fig. 7. Qualitative comparison on a full-resolution sample from GF2 dataset. The first row displays RGB visualizations, while the second row presents the corresponding HQNR maps. Darker colors indicate better pan-sharpening performance.

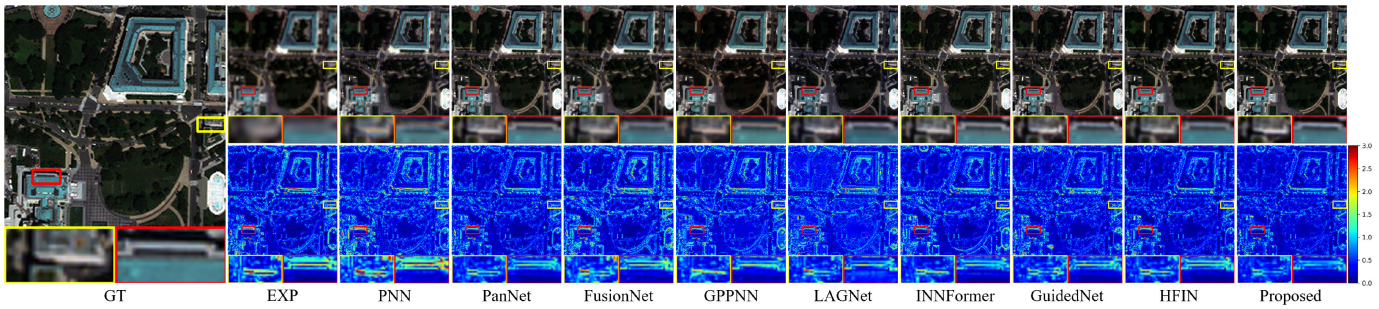


Fig. 8. Qualitative comparison on a reduced-resolution sample from WV2 dataset. The first row shows RGB visualizations, while the second row gives the corresponding absolute error maps between the pan-sharpened results and the GT image.

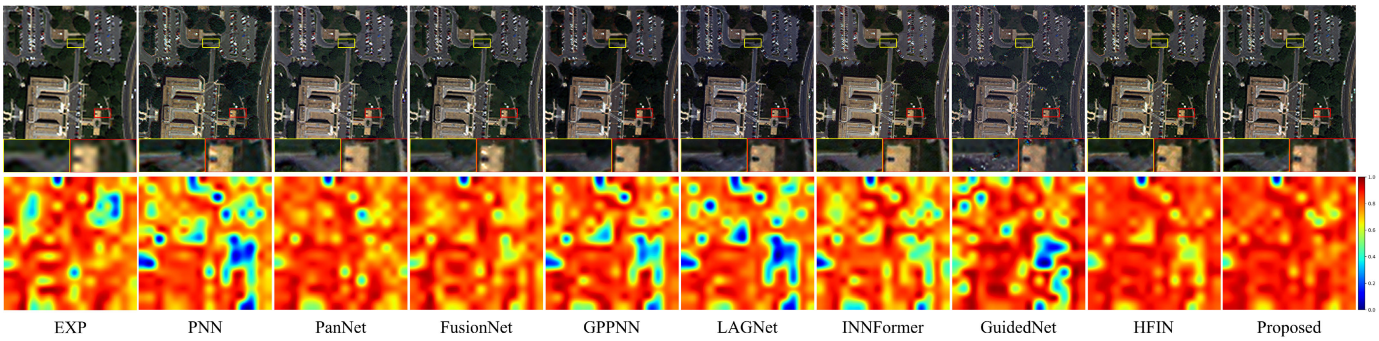


Fig. 9. Qualitative comparison on a full-resolution sample from WV2 dataset. The first row displays RGB visualizations, while the second row presents the corresponding HQNR maps. Darker colors indicate better pan-sharpening performance.

exceptional visual fidelity compared with other DL-based methods. The last row of Figs. 6 and 7 further displays the HQNR maps of the HRMS images generated by our method and other SOTA approaches. Higher HQNR values indicate better overall pan-sharpening performance; thus, the distinctly darker colors in our HQNR map underscore the significant advantage over other methods in generalization performance. This aligns with the trends observed in the quantitative comparisons, further confirming the superiority of our approach. The pleasing performance of our method on the full-resolution demonstrates that our model can be effectively applied to real-world scenes.

3) Generalization Performance Comparison: To further investigate the adaptability and generalization ability of our model, we conduct comparisons between our method and other DL-based approaches on the cross-sensor dataset. To be specific, we use samples from the WV2 sensor to test all DL-based models that were trained on the WV3 dataset without any fine tuning. As clearly shown in Table II, our proposed framework gains the best results for most indexes again, indicating its pleasing applicability and robustness. In particular, the average gains of our method over the second-best PanNet are 0.2239 dB in the reference metric PSNR on the WV2 dataset. The visualized comparisons of reduced-resolution and full-resolution samples shown in Figs. 8 and 9 also demonstrate the favorable cross-sensor performance of our proposed method. In RGB visualizations on the first row of Figs. 8 and 9, our model presents better spatial textures and pleasing spectral preservation. In addition, on the last row of Figs. 8 and 9, the noticeably darker colors in our error map and HQNR map further emphasize the strong generalization capability of our framework.

4) Efficiency-Performance Analysis: Table III reports the efficiency comparisons between our model and other DL-based

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TABLE II
QUANTITATIVE COMPARISON BETWEEN OUR MODEL AND OTHER SOTA DL-BASED METHODS OVER THE REDUCED-RESOLUTION AND FULL-RESOLUTION SAMPLES FROM THE WORLDVIEW-2 DATASET. (RED: BEST AND BLUE: SECOND BEST)

Method	Reduced					Full		
	PSNR↑	SAM↓	ERGAS↓	Q8↑	SCC↑	D_λ ↓	D_s ↓	HQNR↑
EXP	26.4491	6.5323	6.7706	0.6465	0.7357	0.0576	0.0599	0.9040
PNN	28.0445	7.1158	5.6152	0.7619	0.8782	0.1488	0.0771	0.7866
PanNet	30.2761	6.9216	6.2507	0.7205	0.8552	0.1416	0.1023	0.7697
FusionNet	28.7336	6.4257	5.1363	0.7961	0.8746	0.0523	0.0559	0.8944
GPPNN	28.3839	6.8295	5.1743	0.7942	0.9086	0.1194	0.0550	0.8323
LAGNet	28.0495	6.9545	5.3262	0.8054	0.9125	0.1308	0.0547	0.8223
INNFormer	29.8112	6.2793	4.6189	0.8237	0.9170	0.1403	0.0376	0.8276
GuideNet	29.8452	6.2241	4.9145	0.8214	0.8980	0.0545	0.0723	0.8769
HFIN	30.0930	5.4666	4.4107	0.8347	0.9178	0.0586	0.0394	0.9049
Proposed	30.5000	5.6262	4.2210	0.8497	0.9259	0.0419	0.0350	0.9245

TABLE III
EFFICIENCY COMPARISONS WITH OTHER SOTA METHODS EXP: A POLYNOMIAL INTERPOLATION METHOD [26] USED TO UPSAMPLE MS IMAGES

Method	Params(M)	FLOPs(G)	Latency(s)
EXP	-	0.01	-
PNN	0.11	0.30	0.04
PanNet	0.08	5.17	0.02
FusionNet	0.08	0.32	0.07
GPPNN	0.24	0.98	0.02
LAGNet	0.15	0.53	1.38
INNFormer	0.14	5.12	1.33
GuidedNet	1.13	2.29	0.05
HFIN	0.19	8.51	0.28
Proposed	0.35	1.29	0.04

methods. It is evident that, when considered alongside the quantitative experimental results in Tables I and II, our method delivers the best performance while maintaining an acceptable computational cost and favorable inference time, striking a good tradeoff between efficiency and performance.

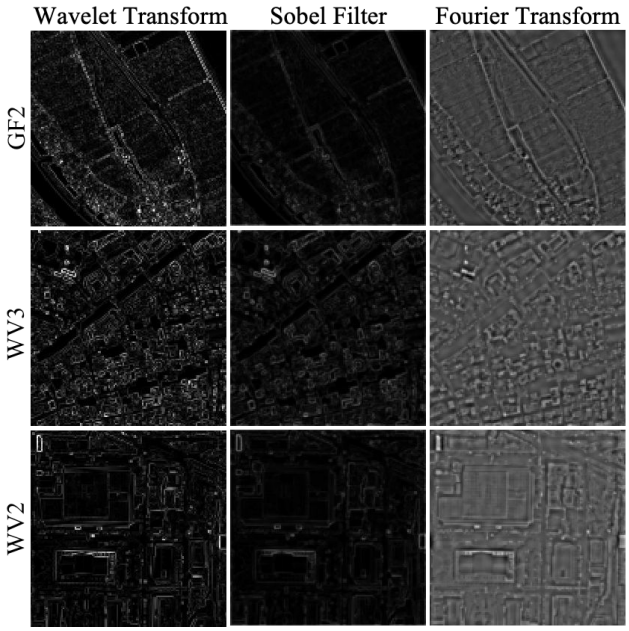


Fig. 10. High-frequency components extracted using different high-frequency extraction techniques.

C. Ablation Experiments

We first verify the effectiveness of the core HFFA and HFDC modules within our method. In addition, we also investigate the effect of different high-frequency extraction manners and coefficients of the loss functions. All ablation experiments are implemented over WV3 satellite dataset.

1) Effect of HFFA and HFDC Modules: We remove the HFFA and HFDC modules separately to verify their contributions. As shown in Configuration (a) of Table IV, removing the HFFA module causes a significant drop in all metrics, especially in full-resolution scenes, underscoring its importance for enhancing the model's generalization capability. In Configuration (b) of Table IV, we delete the HFDC module to assess its necessity. It can be observed that removing the HFDC leads to a notable decline in all metrics as well, demonstrating its efficacy.

2) Effect of the High-frequency Extraction Manners: In this work, we utilize wavelet transform for high-frequency feature extraction. To evaluate the effectiveness of different high-frequency extraction methods, we introduce two additional commonly used approaches, i.e., Fourier Transform and Sobel filter, as replacements for the wavelet-based method, while keeping the other components unchanged. Technically, the first method transforms the feature maps into Fourier space, and then obtains the high-frequency components by filtering out the low-frequency components at the center of the Fourier spectrum [73]; while the latter directly applies the Sobel filter to extract the high-frequency details [74]. As illustrated in Fig. 10, the high-frequency components extracted using wavelet Transform exhibit richer and finer-grained detail information compared to those derived from the other two methods. From Table V, our model equipped with wavelet

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TABLE IV
ABLATION EXPERIMENTS OF THE HFFA AND HFDC MODULES ON WV3 DATASET. (RED: BEST)

Config.	HFFA	HFDC	Reduced					Full		
			PSNR↑	SAM↓	ERGAS↓	Q8↑	SCC↑	D_λ ↓	D_s ↓	HQNR↑
(a)	×	✓	38.4534	2.9765	2.1739	0.9182	0.9850	0.0263	0.0443	0.9306
(b)	✓	×	38.5709	3.0554	2.3056	0.9143	0.9837	0.0212	0.0347	0.9448
(c)	✓	✓	39.0558	2.9579	2.1706	0.9183	0.9858	0.0175	0.0344	0.9487

TABLE V
COMPARISONS OF DIFFERENT HIGH-FREQUENCY EXTRACTION MANNERS ON WV3 DATASET. (RED: BEST)

Config.	Reduced					Full		
	PSNR↑	SAM↓	ERGAS↓	Q8↑	SCC↑	D_λ ↓	D_s ↓	HQNR↑
Fourier Transform	38.3456	3.0085	2.1964	0.9168	0.9798	0.0240	0.0419	0.9353
Sobel Filter	38.5632	2.9783	2.1796	0.9169	0.9801	0.0204	0.0354	0.9449
Wavelet Transform	39.0558	2.9579	2.1706	0.9183	0.9858	0.0175	0.0344	0.9487

TABLE VI
ABLATION STUDIES OF THE LOSS FUNCTION ON WV3 DATASET. (RED: BEST)

Config.	β	γ	Reduced					Full		
			PSNR↑	SAM↓	ERGAS↓	Q8↑	SCC↑	D_λ ↓	D_s ↓	HQNR↑
(a)	0.5	0.05	38.1202	3.0264	2.1878	0.9172	0.9842	0.0267	0.0408	0.9336
(b)	0.5	0.1	38.3299	2.9735	2.1720	0.9182	0.9832	0.0234	0.0436	0.9340
(c)	1	0.05	39.0028	2.9680	2.1753	0.9176	0.9847	0.0201	0.0392	0.9415
(d)	1	0.1	39.0558	2.9579	2.1706	0.9183	0.9858	0.0175	0.0344	0.9487

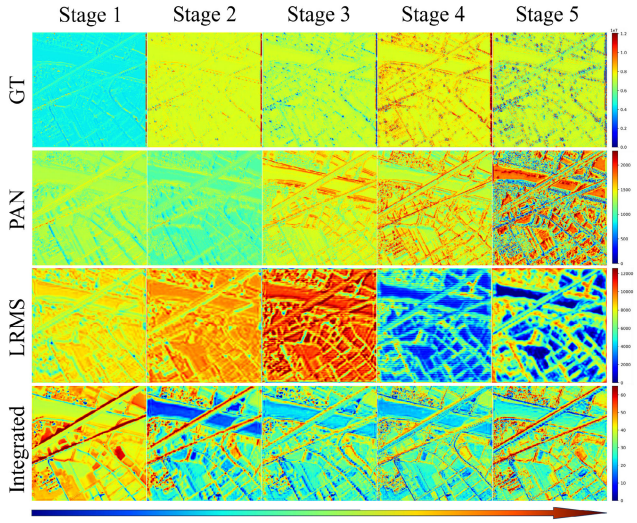


Fig. 11. Visualization of feature maps across different stages. The first three rows represent the features of the high-frequency components of LRMS, PAN, and GT images, respectively, while the last row indicates the features of the HFII module. It can be seen that the feature maps of the HFII module gradually exhibit clear and rich textures as the stage number increases, demonstrating the progressive restoration of high-frequency details.

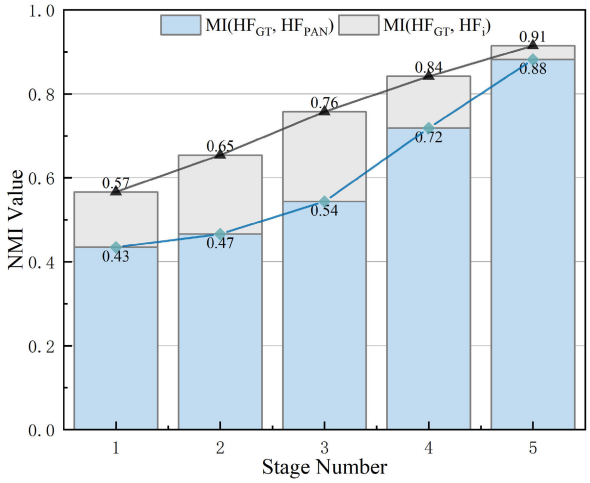


Fig. 12. Variation of the mean NMI in the high-frequency across different stages. Blue line: NMI between GT features (HF_{GT}) and PAN features (HF_{PAN}). The gray line: the NMI between GT features (HF_{GT}) and the integrated features (HF_i) from the HFII module. It is clear that both NMI values increase as the stage progresses, suggesting that the high-frequency components of PAN and the integrated results increasingly align with those of GT. In other words, high-frequency details are precisely attained via our cooperative optimization strategy.

Transform yields the best results across both the reduced- and full-resolution metrics.

3) Effect of the Loss Function Coefficients: We assess the impact of different weighting coefficients for our core loss terms L_{MI} and L_{CL} in the total loss function $L = L_{rec} + \alpha L_{SSIM} + \beta L_{MI} + \gamma L_{CL}$. Specifically, we vary their weighting parameters, β and γ, respectively, while keeping the coefficient α of L_{SSIM} fixed at 0.1. The results, displayed in Table VI, indicate that the optimal settings are $\beta = 1$ and $\gamma = 0.1$.

D. Visualization of Feature Maps

To further validate the capability of our proposed framework, we present high-frequency feature maps at different

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stages in Fig. 11. As evidently shown, with the stage number increasing, the texture information becomes progressively richer and clearer. The high-frequency details of PAN feature gradually emerge under the guidance of GT features, demonstrating the efficacy of the MI maximization constraint in the HFFA module. Simultaneously, the previously blurred edges of LRMS features are sharpened via our devised contrastive learning constraints. Ultimately, our integrated features successfully capture both the fine-grained details and favorable spectral preservation from different modalities, thereby affirming the effectiveness of our proposed cooperative optimization strategy. In addition, the normalized MI (NMI) statistics, displayed in Fig. 12, increase with the stage number for both GT/PAN and GT/integrated results. The higher NMI of the integrated results compared with PAN indicates efficient multimodal high-frequency learning through feature integration.

V. Conclusion

In this article, we propose a general framework tailored for addressing the challenging high-frequency reconstruction in pan-sharpening learning. Specifically, our method features two fundamental procedures, i.e., the high-frequency alignment and the high-frequency learning which are implemented through the key HFFA module and HFDC module, respectively. Technically, the HFFA module maximizes the MI between the high-frequency features of PAN images and GT images, thereby encouraging those of PAN to align with GT. The HFDC module calibrates inaccurate LRMS components in cross-modal high-frequency features with contrastive learning constraints, resulting in more precise cross-modal detail formation. Our model, built upon these two core designs, can generate images with rich spatial details and desirable spectral preservation. Extensive experiments on various satellite datasets demonstrate its superiority. While our method achieves optimal performance, we aim to further investigate its adaptability and applicability by applying the key cooperative optimization strategy to various pan-sharpening architectures, as well as to other image fusion tasks, such as infrared and visible image fusion, hyperspectral, and multispectral image fusion.

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